Lynea Database Relationships

Purpose: This document explains how Lynea's proposed PostgreSQL tables connect, what each relationship means, and how to represent those relationships in an ER diagram.

Relationship Summary

Relationship	Cardinality	Meaning
users → trip_intents	1 to many	One user can submit multiple travel requests over time. Each request belongs to one user.
users → group_members	1 to many	A user can join multiple groups. Each membership row refers to one user.
travel_groups → group_members	1 to many	A group can have many members. Each membership belongs to one group.
group_members → trip_preferences	1 to 0..1	A membership can have zero or one preference record initially. UNIQUE on group_member_id enforces at most one.
trip_preferences → member_interests	1 to many	One preference record can contain several selected interests.
interests → member_interests	1 to many	An interest can be selected by many preference records.
travel_groups → recommendation_runs	1 to many	A group can request recommendations repeatedly as preferences change.
recommendation_runs → trip_options	1 to many	One AI run can produce multiple destination options.
trip_options → votes	1 to many	Each option can receive votes from multiple group members.
group_members → votes	1 to many	Each member can vote on multiple options.

Users and travel_groups have a many-to-many relationship. One person may participate in several travel groups, and each group may contain many people. Relational databases represent this using the group_members junction table.:

- User A is a member of Group 1.
- User B is a member of Group 1.
- User A is also a member of Group 2.

These are three rows in group_members. The UNIQUE (group_id, user_id) constraint prevents a user from being added to the same group more than once. trip_preferences and interests are also many-to-many. A single person may select multiple interests, and an interest such as hiking may be selected by many people. The member_interests junction table stores each selection and its priority. Its composite primary key is (preference_id, interest_id).

Text ER Diagram

USERS 1 ——< TRIP_INTENTS

USERS 1 ——< GROUP_MEMBERS >—— 1 TRAVEL_GROUPS

GROUP_MEMBERS 1 —— 0..1 TRIP_PREFERENCES

TRIP_PREFERENCES 1 ——< MEMBER_INTERESTS >—— 1 INTERESTS

TRAVEL_GROUPS 1 ——< RECOMMENDATION_RUNS

RECOMMENDATION_RUNS 1 ——< TRIP_OPTIONS

TRIP_OPTIONS 1 ——< VOTES >—— 1 GROUP_MEMBERS

1 means one record; < means many records; 0..1 means optional, at most one. The group_members table resolves the many-to-many relationship between users and travel_groups. The member_interests table resolves the many-to-many relationship between trip_preferences and interests.

Foreign Keys and Constraints

Foreign-key column	References	Rule
trip_intents.user_id	users.user_id	Every travel request must belong to an existing user.
group_members.user_id	users.user_id	Every membership must reference an existing user.
group_members.group_id	travel_groups.group_id	Every membership must reference an existing group.
trip_preferences.group_member_id	group_members.group_member_id	Preferences are tied to a specific group membership.
member_interests.preference_id	trip_preferences.preference_id	Interest selections must reference an existing preference record.
member_interests.interest_id	interests.interest_id	Interest selections must reference an existing interest.
recommendation_runs.group_id	travel_groups.group_id	Each AI generation belongs to an existing group.
trip_options.run_id	recommendation_runs.run_id	Each option must belong to a recommendation run.
votes.option_id	trip_options.option_id	Each vote must refer to an existing option.
votes.group_member_id	group_members.group_member_id	Each vote must come from an existing group membership.